

Discussion 10 Worksheet

DS 120 Staff

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Formula Bank

- **Permutations:** The number of ways to arrange r objects from n distinct items:

$$P(n, r) = \frac{n!}{(n-r)!}$$

- **Combinations:** The number of ways to choose r objects from n distinct items (order does not matter):

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- **Binomial Formula:** For any integers $n \geq 0$,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

- **Binomial Probability Formula:** If $X \sim \text{Binomial}(n, p)$, then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Warm-Up

1. Compute the number of ways to arrange 5 books on a shelf if you only select 3 of them.
2. A committee of 4 is chosen from 10 students. How many different committees are possible?

Permutations and Combinations

Assume all of these problems operate without replacement.

1. Investment firms usually have a large selection of mutual funds from which an investor can choose. One such firm has 30 mutual funds. Suppose that you plan to invest in 4 of these mutual funds, 1 during each quarter of next year. Use permutation notation to express the number of possibilities and then evaluate that expression.
2. The IRS decides that it will audit the returns of 3 people from a group of 18. Use combination notation to express the number of possibilities and then evaluate that expression.
3. The powerball is a multistate lottery that was introduced in April 1992, and is now sold in 24 states, the District of Columbia, and the Virgin Islands. To play the game, a player first selects five numbers from the numbers 1-53 and then chooses a Powerball number, which can be any number between 1 and 42, inclusive. How many possibilities are there?
4. The US Senate consists of 100 senators, 2 from each state. A committee consisting of 5 senators is to be formed. How many different committees are possible? How many are possible if no state can have more than 1 senator on the committee?
5. How many distinct arrangements are there of the letters in the word PERSEVERE?

Binomial Probability Formula

- **Bernoulli Trials:** Repeated trials of a random experiment are called **Bernoulli Trials** if the following conditions hold:
 1. The trials are independent of one another
 2. The result of each trial is classified as a success or failure, depending on whether or not a specified event occurs, respectively.
 3. The success probability – and hence the failure probability – remains the same from trial to trial
- **Binomial Random Variables:** Given a sequence of n Bernoulli Trials with success probability p , the random variable X of a Binomial RV denotes the number of successes in n trials.

Binomial Practice Problems:

1. According to the *Daily Racing Form*, the probability is about 0.67 that the favorite in a horse race will finish in the money (first, second, or third place). In the next five races, what is the probability that the favorite finishes in the money
 - (a) exactly twice?
 - (b) exactly four times?
 - (c) at least four times?
 - (d) between two and four times, inclusive?
2. If all sex distributions are equally likely, what proportion of families with five children have three girls and two boys?

3. A baseball player has a batting average of 0.260. Suppose that you observe successive at-bats of the player and note for each at bat whether the player gets a hit
- (a) Under what conditions is the assumption of Bernoulli Trials appropriate?
 - (b) Assuming your conditions above, what is the probability that the player will get two or more hits in his next four times at-bat?